Chapter 13 · Semester 2

ELECTROSTATICS

Notes by Cikgu Suhaili

23 worked examples 3 simulations Exercises with answers

Electrostatics is the study of **electric charges at rest**, the forces between them and the electric fields around them.

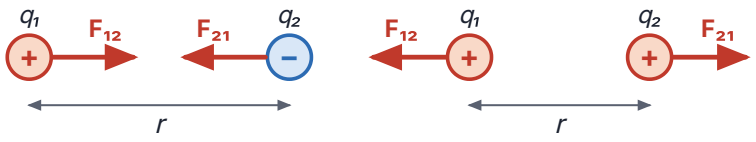
13.1 Coulomb's law

Coulomb's law

The electric force between two point charges is **directly proportional to the product of their charges** and **inversely proportional to the square of the distance** between them.

$$F = \frac{Qq}{4\pi\varepsilon_0 r^2} = \frac{kQq}{r^2}$$

$k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9\ \text{N m}^2\ \text{C}^{-2}$ (Coulomb's constant); $\varepsilon_0 = 8.85 \times 10^{-12}\ \text{C}^2\ \text{N}^{-1}\ \text{m}^{-2}$ (permittivity of free space)



unlike charges attract like charges repel

F_{12} = force on q_1 due to q_2; F_{21} = force on q_2 due to q_1

$F_{12} = -F_{21}$ (Newton's third law)

Figure 13.1: the forces between two point charges are equal in size and opposite in direction.

- Like charges repel; unlike charges attract.
- The two forces form an action–reaction pair: $F_{12} = F_{21} = kq_1q_2/r^2$, even if the charges are different in size.
- Doubling r makes the force four times smaller (inverse-square law).



Figure 13.2: F against r (a curve) and F against $1/r^2$ (a straight line through the origin).

Teacher's note

Use only the **size** of each charge in Coulomb's law. Decide the **direction** separately from "like repel, unlike attract", then draw the force on the diagram.

Relative permittivity

When the charges are in a medium (not a vacuum), the force is smaller. Replace ε_0 by the permittivity of the medium, ε.

Relative permittivity, ε_r

the **ratio of the permittivity of a medium to the permittivity of free space**. It has no unit.

$$\varepsilon_r = \frac{\varepsilon}{\varepsilon_0} \qquad F_{\text{medium}} = \frac{Qq}{4\pi\varepsilon_r\varepsilon_0 r^2} = \frac{F_{\text{vacuum}}}{\varepsilon_r}$$

ε_r of air ≈ 1.0006 (taken as 1); water ≈ 80; paper ≈ 3.7. The bigger ε_r, the weaker the force between the charges.

More than two charges

The resultant force on one charge is the **vector sum** of the forces from each of the other charges. Work out each force separately, then add them as vectors (along a line: add with signs; in two dimensions: resolve into x and y components).

Example 13.1.1

Two point charges q_1 = −20 nC and q_2 = +90 nC are 4.0 cm apart. Find the magnitude and direction of (a) the electric force on q_1 due to q_2, (b) the electric force on q_2 due to q_1. (k = 9.0 × 10^9 N m^2 C^{-2})



Solution

(a) $F_{12} = \dfrac{kq_1q_2}{r^2} = \dfrac{(9.0 \times 10^9)(2.0 \times 10^{-8})(9.0 \times 10^{-8})}{(4.0 \times 10^{-2})^2}$ = **1.0 × 10^{-2} N**, towards q_2 (attraction)

(b) F_{21} = **1.0 × 10^{-2} N**, towards q_1: equal in size and opposite in direction (Newton's third law)

Example 13.1.2

Three point charges lie on the x-axis: q_1 = +2 μC, q_2 = +4 μC (3.0 cm to the right of q_1) and q_3 = −6 μC (5.0 cm to the right of q_2). Find the net electric force on q_2.



Solution

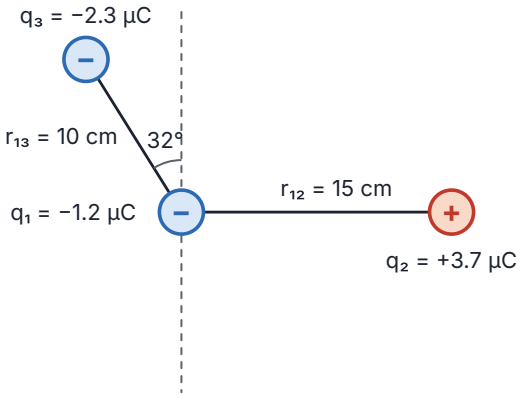
$F_{21} = \dfrac{(9.0 \times 10^9)(4 \times 10^{-6})(2 \times 10^{-6})}{(3.0 \times 10^{-2})^2}$ = 80 N, to the right (q_1 repels q_2)

$F_{23} = \dfrac{(9.0 \times 10^9)(4 \times 10^{-6})(6 \times 10^{-6})}{(5.0 \times 10^{-2})^2}$ = 86.4 N, to the right (q_3 attracts q_2)

Both forces point the same way: F_2 = 80 + 86.4 = **166 N, to the right** (towards q_3)

Example 13.1.3

Three point charges are placed as shown. q_1 = −1.2 μC, q_2 = +3.7 μC, q_3 = −2.3 μC, r_{12} = 15 cm, r_{13} = 10 cm and the line q_1q_3 makes 32° with the vertical. Find the magnitude and direction of the resultant force on q_1.



Solution

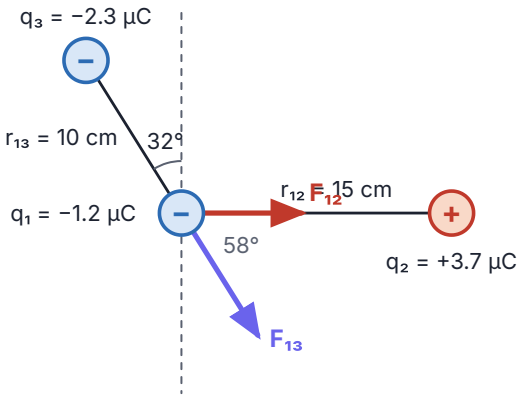
$F_{12} = \dfrac{(9.0 \times 10^9)(1.2 \times 10^{-6})(3.7 \times 10^{-6})}{(0.15)^2}$ = 1.78 N, along $+x$ (attracted towards q_2)

$F_{13} = \dfrac{(9.0 \times 10^9)(1.2 \times 10^{-6})(2.3 \times 10^{-6})}{(0.10)^2}$ = 2.48 N, away from q_3 (repulsion): down and to the right at 58° below the horizontal

ΣF_x = 1.78 + 2.48 cos 58° = 1.78 + 1.31 = 3.09 N

ΣF_y = −2.48 sin 58° = −2.10 N

$F_1 = \sqrt{3.09^2 + 2.10^2}$ = **3.74 N**; θ = $\tan^{-1}(2.10/3.09)$ = **34.2° below the $+x$ direction**



Example 13.1.4

Two charges +3.0 μC and −2.0 μC are 5.0 cm apart in oil of relative permittivity 2.5. Find the force between them.

Solution

$$F = \frac{Qq}{4\pi\varepsilon_r\varepsilon_0 r^2} = \frac{(9.0\times10^{9})(3.0\times10^{-6})(2.0\times10^{-6})}{2.5\times(0.050)^2} = \textbf{8.6 N}, \text{ attractive}$$

In air the force would be 21.6 N; the oil reduces it by a factor of $\varepsilon_r = 2.5$.

Exercise: Coulomb's law

1. Two point charges of -1.0×10^{-6} C and $+2.0\times10^{-6}$ C are 0.30 m apart. What is the electrostatic force on each charge?

   Answer

   0.20 N, each directed towards the other

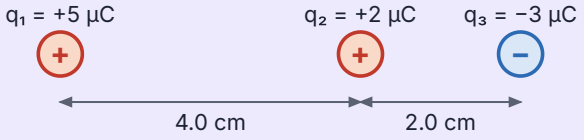
   FULL SOLUTION

   $F = kq_1q_2/r^2 = (9.0\times10^{9})(1.0\times10^{-6})(2.0\times10^{-6}) / (0.30)^2$

   $= \mathbf{0.20\ N}$

   The charges are opposite, so the force is attractive: **each force is directed towards the other charge** (Newton's third law: equal and opposite).

2. Calculate the net electrostatic force on (i) q_1, (ii) q_2, (iii) q_3.



Answer

(i) 18.75 N to the left (ii) 191.25 N to the right (iii) 172.5 N to the left

FULL SOLUTION

$q_1 = +5\ \mu C$ at x = 0, $q_2 = +2\ \mu C$ at 4.0 cm, $q_3 = -3\ \mu C$ at 6.0 cm.

$F_{12} = (9.0 \times 10^9)(5 \times 10^{-6})(2 \times 10^{-6}) / (0.040)^2 = 56.25$ N (repulsive)

$F_{13} = (9.0 \times 10^9)(5 \times 10^{-6})(3 \times 10^{-6}) / (0.060)^2 = 37.5$ N (attractive)

$F_{23} = (9.0 \times 10^9)(2 \times 10^{-6})(3 \times 10^{-6}) / (0.020)^2 = 135$ N (attractive)

(i) On q_1: q_2 repels it to the left (56.25 N), q_3 attracts it to the right (37.5 N)

F = 56.25 − 37.5 = **18.75 N to the left**

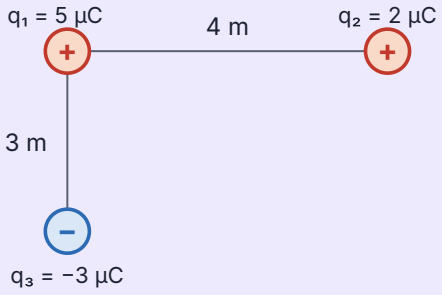
(ii) On q_2: q_1 repels it to the right (56.25 N), q_3 attracts it to the right (135 N)

F = 56.25 + 135 = **191.25 N to the right**

(iii) On q_3: q_1 and q_2 both attract it to the left

F = 37.5 + 135 = **172.5 N to the left**

3. Calculate the net electrostatic force on q_2.



Answer

4.11×10^{-3} N, 18.4° below the +x direction

FULL SOLUTION

Place q_1 at the origin, q_2 at (4 m, 0) and q_3 at (0, −3 m). Distance q_2 to $q_3 = \sqrt{(4^2 + 3^2)} = 5$ m.

$F_{12} = (9.0 \times 10^9)(5 \times 10^{-6})(2 \times 10^{-6}) / 4^2 = 5.625 \times 10^{-3}$ N, repulsive, in the +x direction

$F_{32} = (9.0 \times 10^9)(3 \times 10^{-6})(2 \times 10^{-6}) / 5^2 = 2.16 \times 10^{-3}$ N, attractive, towards q_3, along (−4, −3)/5

$F_x = 5.625 \times 10^{-3} - 2.16 \times 10^{-3}(0.8) = 3.897 \times 10^{-3}$ N

$F_y = -2.16 \times 10^{-3}(0.6) = -1.296 \times 10^{-3}$ N

$F = \sqrt{(F_x^2 + F_y^2)}$

$= \mathbf{4.11 \times 10^{-3}\ N}$

tan θ = 1.296/3.897 ⇒ θ = **18.4° below the +x direction**

4. $q_1 = +2.5\ \mu C$ is at (0, +0.30 m), $q_2 = +2.5\ \mu C$ is at (0, −0.30 m) and $q_3 = +3\ \mu C$ is at (+0.40 m, 0). What is the electrostatic force on q_3?

Answer

0.43 N in the $+x$ direction (each force is 0.27 N at 0.50 m; the y components cancel)

FULL SOLUTION

Distance of q_1 (and of q_2) from q_3: $r = \sqrt{(0.30^2 + 0.40^2)} = 0.50$ m

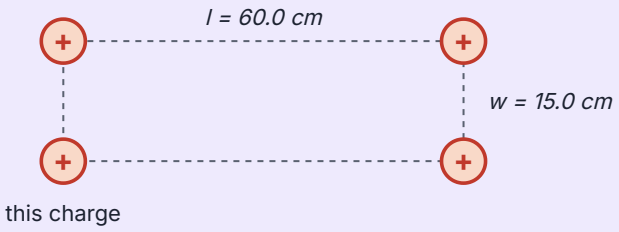
Each force: $F = (9.0 \times 10^9)(2.5 \times 10^{-6})(3 \times 10^{-6}) / (0.50)^2 = 0.27$ N (repulsive)

By symmetry the y components cancel; each x component = $0.27 \times (0.40/0.50) = 0.216$ N

$F = 2 \times 0.216$

= **0.43 N in the +x direction**

5. Four identical point charges (q = +10.0 μC) are at the corners of a rectangle, l = 60.0 cm and w = 15.0 cm. Find the magnitude and direction of the resultant force on the charge at the lower left corner.



Answer

40.9 N at 263° (measured anticlockwise from the $+x$ direction): almost straight down

FULL SOLUTION

Put the lower-left charge at the origin; the others are at (0.600, 0), (0, 0.150) and (0.600, 0.150) m. All forces are repulsive. $kq^2 = (9.0 \times 10^9)(10.0 \times 10^{-6})^2 = 0.90$ N m^2.

Lower-right: $F = 0.90 / 0.600^2 = 2.50$ N in the −x direction

Upper-left: $F = 0.90 / 0.150^2 = 40.0$ N in the −y direction

Upper-right: $r^2 = 0.600^2 + 0.150^2 = 0.3825$ m^2, $F = 0.90/0.3825 = 2.35$ N along (−0.600, −0.150)/0.6185

components: (−2.28, −0.571) N

$\Sigma F_x = -2.50 - 2.28 = -4.78$ N; $\Sigma F_y = -40.0 - 0.571 = -40.57$ N

$F = \sqrt{(4.78^2 + 40.57^2)}$

= **40.9 N**

$\tan \varphi = 40.57/4.78 \Rightarrow \varphi = 83.3°$ below the −x direction, i.e. **263° anticlockwise from +x** (almost straight down)

13.2 Electric field

> **Electric field**
>
> a region around a charge where another charge experiences an electric force.

An electric field is shown with **electric field lines** (lines of force). The direction of the field at a point is the tangent to the line at that point: the direction of the force on a **positive** test charge.

Field patterns

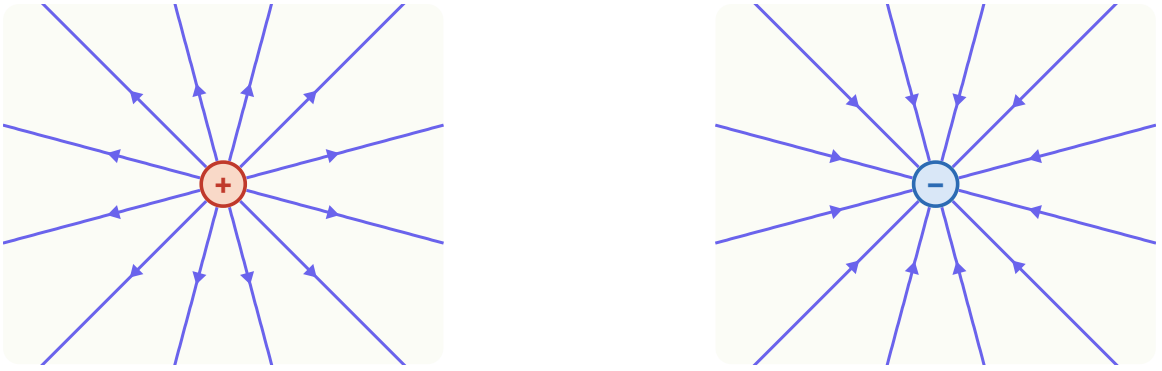


Figure 13.3: an isolated point charge. Positive: the lines point radially outwards. Negative: the lines point radially inwards.

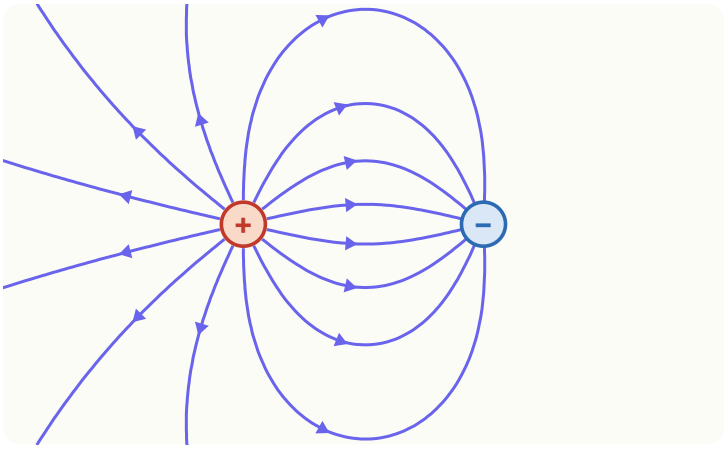


Figure 13.4: an electric dipole ($+q$ and $-q$). The lines curve from the positive charge to the negative charge.

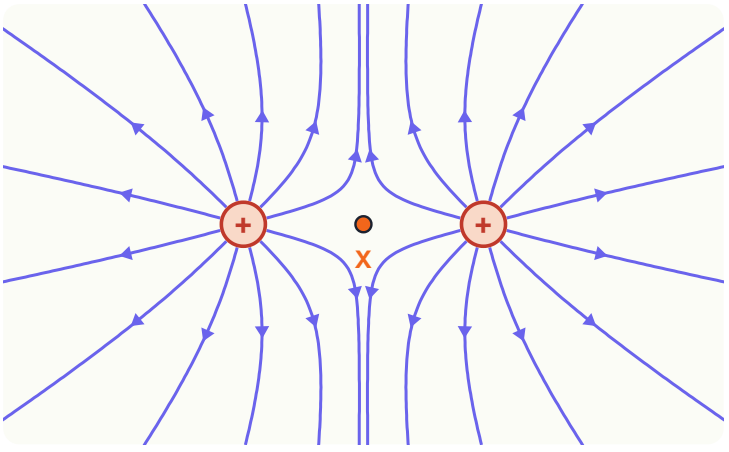


Figure 13.5: two equal positive charges. X is the **neutral point**: the resultant field (and force on a charge) there is zero.

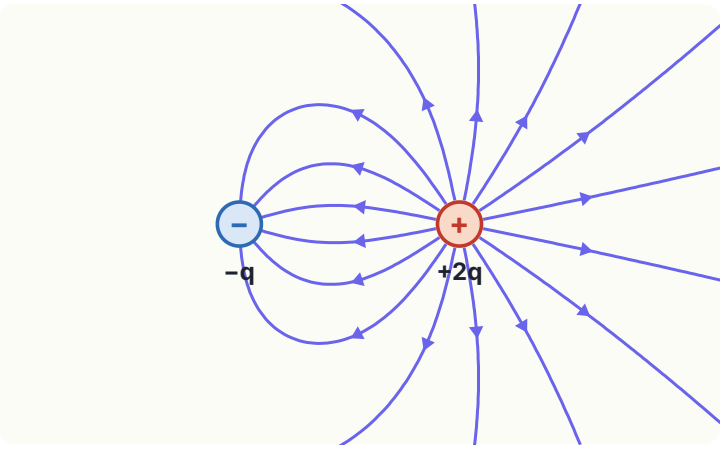
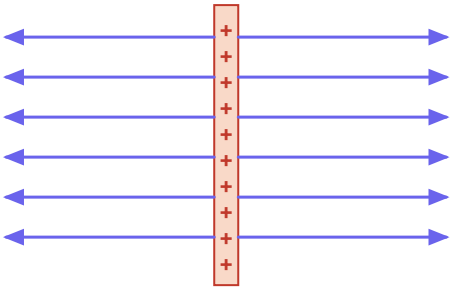
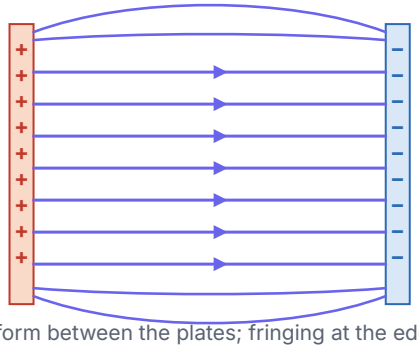


Figure 13.6: $+2q$ and $-q$. Twice as many lines leave $+2q$ as enter $-q$; the rest go off to infinity.



field lines perpendicular to the surface, on both sides

Figure 13.7: a uniformly charged surface (plate).



uniform between the plates; fringing at the edges

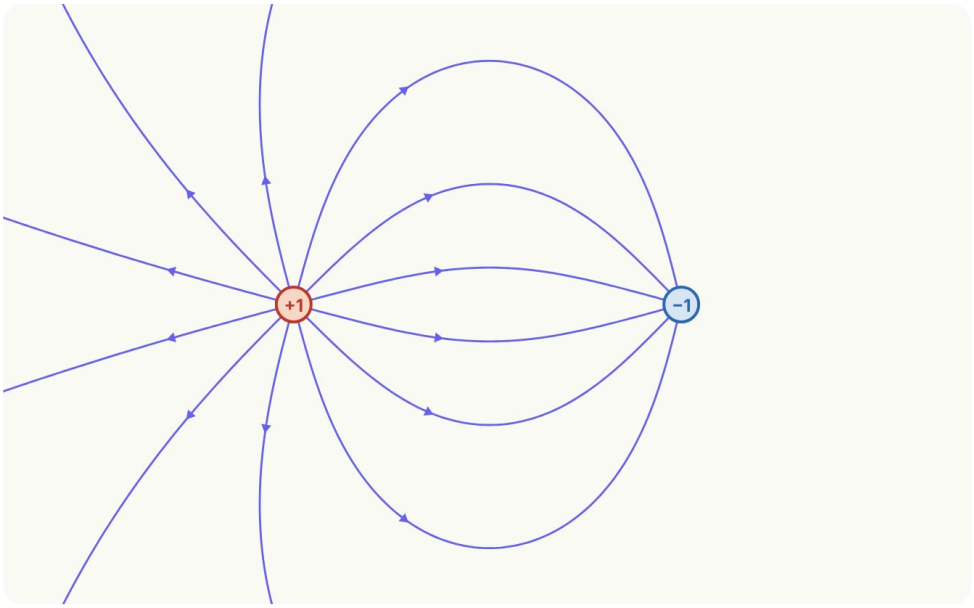
Figure 13.8: two oppositely charged parallel plates.

- Field lines start on positive charges and end on negative charges. The number of lines is proportional to the size of the charge.
- The closer the lines, the stronger the field.
- Field lines **never cross**: the field cannot have two directions at one point.
- At a charged conducting surface, the lines are **perpendicular** to the surface.
- Between two parallel plates the lines are parallel and equally spaced: the field is **uniform**. The fringing at the edges is ignored when the separation is small compared with the size of the plates.

Simulation

Electric field lines and the field at a point

Choose a set of charges and change the second charge. Tap anywhere to place a test point and read the field strength and potential there. Turn on the equipotentials to see that they always cross the field lines at right angles.



tap the picture	scale	charges
to place a test point	**picture width = 40 cm**	**q1 = +1 µC, q2 = −1 µC**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 13 to use the simulation.

Electric field strength

Electric field strength, E

the **electric force per unit positive charge** acting on a small test charge placed at that point.

$$E = \frac{F}{q} \qquad \text{for a point charge: } E = \frac{Q}{4\pi\varepsilon_0 r^2} = \frac{kQ}{r^2}$$

vector; unit N C^{-1} (= V m^{-1}). Points away from a positive charge and towards a negative charge.

Derivation for a point charge: a test charge q at distance r from Q feels $F = kQq/r^2$, so $E = F/q = kQ/r^2$. It does not depend on the test charge.

Teacher's note

For several charges, find E from each charge using only its size, mark its direction on the diagram (away from +, towards −), then add the fields as **vectors**. The force on a charge q placed there is $F = qE$ (opposite to E if q is negative).

Example 13.2.1

An electric field of magnitude 0.370 N C^{-1} exerts a force of 6.20 × 10^{-4} N on a test charge. What is the size of the test charge?

Solution

$$q = \frac{F}{E} = \frac{6.20 \times 10^{-4}}{0.370} = \mathbf{1.68 \times 10^{-3}\ C}\ (1.68\ \text{mC})$$

Example 13.2.2

Find the electric field strength at point A, 2.0 cm from a point charge of (a) +5.0 μC, (b) −5.0 μC.



Solution

$$E = \frac{kq}{r^2} = \frac{(9.0 \times 10^{9})(5.0 \times 10^{-6})}{(2.0 \times 10^{-2})^2} = \mathbf{1.1 \times 10^{8}\ N\ C^{-1}}$$

(a) directed **away from** the charge; (b) the same size, directed **towards** the charge

Example 13.2.3

Point A is 2.0 cm from q_1 = +1.0 μC and 3.0 cm from q_2 = −4.0 μC, on the line between them. Find (a) the electric field at A, (b) the force on a charge q_0 = +4.0 μC placed at A.



Solution

$E_1 = \dfrac{(9.0 \times 10^9)(1.0 \times 10^{-6})}{(0.020)^2} = 2.25 \times 10^7$ N C^{-1}, to the right (away from $+q_1$)

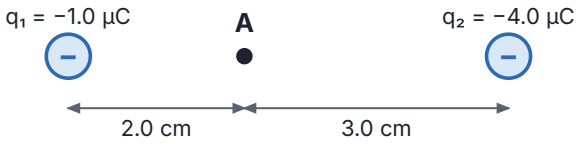
$E_2 = \dfrac{(9.0 \times 10^9)(4.0 \times 10^{-6})}{(0.030)^2} = 4.0 \times 10^7$ N C^{-1}, to the right (towards $-q_2$)

(a) $E = 2.25 \times 10^7 + 4.0 \times 10^7$ = **6.3×10^7 N C^{-1}, to the right**

(b) $F = q_0 E = (4.0 \times 10^{-6})(6.25 \times 10^7)$ = **250 N, to the right**

Example 13.2.4

Repeat Example 13.2.3 with q_1 = −1.0 μC.



Solution

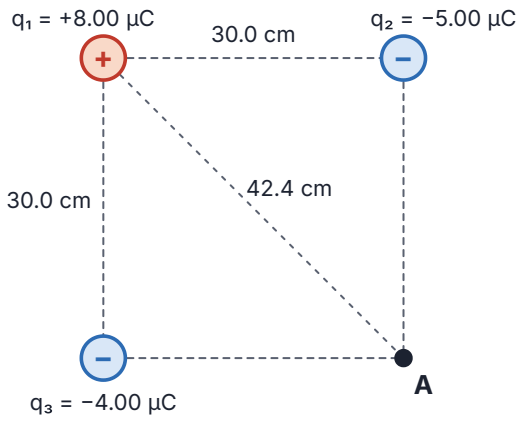
$E_1 = 2.25 \times 10^7$ N C^{-1}, now to the **left** (towards $-q_1$); $E_2 = 4.0 \times 10^7$ N C^{-1}, to the right

(a) $E = 4.0 \times 10^7 - 2.25 \times 10^7$ = **1.75×10^7 N C^{-1}, to the right**

(b) $F = (4.0 \times 10^{-6})(1.75 \times 10^7)$ = **70 N, to the right**

Example 13.2.5

Three charges are at three corners of a square of side 30.0 cm. Find the electric field at the fourth corner A, and the force on a +6.00 μC charge placed at A.



Solution

$E_1 = \dfrac{(9.0 \times 10^9)(8.00 \times 10^{-6})}{(0.424)^2} = 4.00 \times 10^5$ N C^{-1}, away from q_1 (down-right, 45°)

$E_2 = \dfrac{(9.0 \times 10^9)(5.00 \times 10^{-6})}{(0.300)^2} = 5.00 \times 10^5$ N C^{-1}, up (towards q_2)

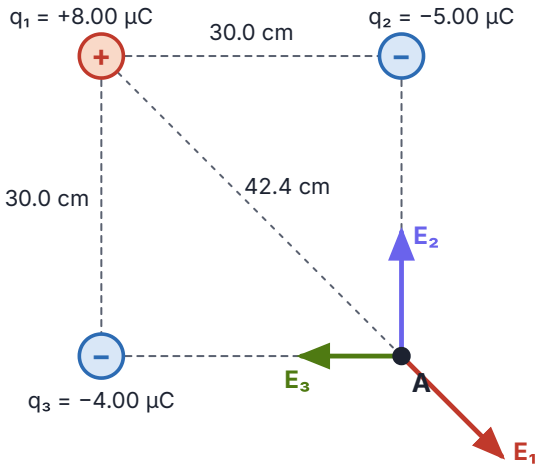
$E_3 = \dfrac{(9.0 \times 10^9)(4.00 \times 10^{-6})}{(0.300)^2} = 4.00 \times 10^5$ N C^{-1}, left (towards q_3)

$\Sigma E_x = 4.00 \times 10^5 \cos 45° - 4.00 \times 10^5 = -1.17 \times 10^5$ N C^{-1}

$\Sigma E_y = 5.00 \times 10^5 - 4.00 \times 10^5 \sin 45° = +2.17 \times 10^5$ N C^{-1}

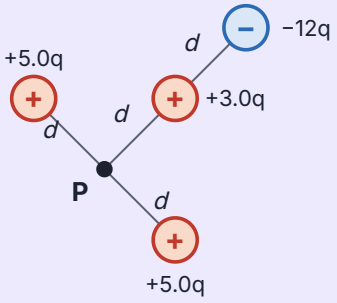
$E = \sqrt{(1.17 \times 10^5)^2 + (2.17 \times 10^5)^2} =$ **2.47×10^5 N C^{-1}**, at $\tan^{-1}(2.17/1.17) =$ **61.7° above the $-x$ direction** (up and to the left)

$F = qE = (6.00 \times 10^{-6})(2.47 \times 10^5) =$ **1.48 N**, in the same direction as E



Exercise: electric field

1. Find the magnitude of the electric field at P due to the four point charges shown, if q = 1 nC and d = 1 cm. ($\varepsilon_0 = 8.85 \times 10^{-12}\ C^2\ N^{-1}\ m^{-2}$)



Answer

Zero. The two +5.0q fields cancel. On the other line, +3.0q at d gives $3kq/d^2$ away from it and −12q at 2d gives $12kq/4d^2 = 3kq/d^2$ towards it: they cancel too.

FULL SOLUTION

$E = kQ/r^2$ for each charge; compare the charges in pairs on each line through P.

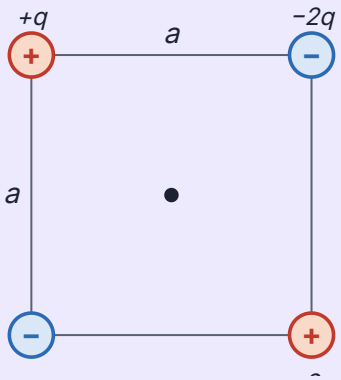
The two +5.0q charges are both at distance d from P on opposite sides: their fields are equal and opposite, so they cancel.

On the other line: +3.0q at distance d gives $E = k(3.0q)/d^2 = 3kq/d^2$, pointing away from it.

−12q at distance 2d gives $E = k(12q)/(2d)^2 = 3kq/d^2$, pointing towards it, i.e. in the opposite direction.

These also cancel, so E at P = **zero**.

2. Find the magnitude and direction of the electric field at the centre of the square if q = 1.0 × 10^{-8} C and a = 5.0 cm.



Answer

1.02 × 10^5 N C^{-1}, upwards

FULL SOLUTION

Distance from each corner to the centre: r = a/√2 = 0.050/√2 = 0.0354 m

For charge q: E_0 = kq/r^2 = (9.0 × 10^9)(1.0 × 10^{-8}) / (0.0354)^2 = 7.2 × 10^4 N C^{-1}

Fields at the centre (away from + charges, towards − charges):

+q (top left): E_0 towards the bottom right

−2q (top right): $2E_0$ towards the top right

−q (bottom left): E_0 towards the bottom left

+2q (bottom right): $2E_0$ towards the top left

Horizontal components: (E_0 + $2E_0$ − E_0 − $2E_0$)/√2 = 0

Vertical components: (−E_0 + $2E_0$ − E_0 + $2E_0$)/√2 = $2E_0$/√2 = √2E_0

E = √2 × 7.2 × 10^4

= **1.02 × 10^5 N C^{-1}, upwards**

Motion of a charge in a uniform electric field

In a uniform field the only force on the charge is $F = qE$ (its weight is negligible). The force is constant, so the acceleration is constant:

$$a = \frac{qE}{m} \qquad E = \frac{V}{d} \text{ between parallel plates}$$

a positive charge accelerates along E; a negative charge (electron) accelerates opposite to E

Moving along the field lines

- The charge speeds up (or slows down) in a straight line, like a body falling under gravity.
- Work done by the field = qV = gain in kinetic energy: $\frac{1}{2}mv^2 = qV$ (from rest)

Entering at right angles to the field

- Horizontal: no force, so $v_x = v_0$ stays constant.
- Vertical: constant acceleration $a = qE/m$.
- The path between the plates is a **parabola** (like a projectile). After leaving the plates it moves in a straight line.

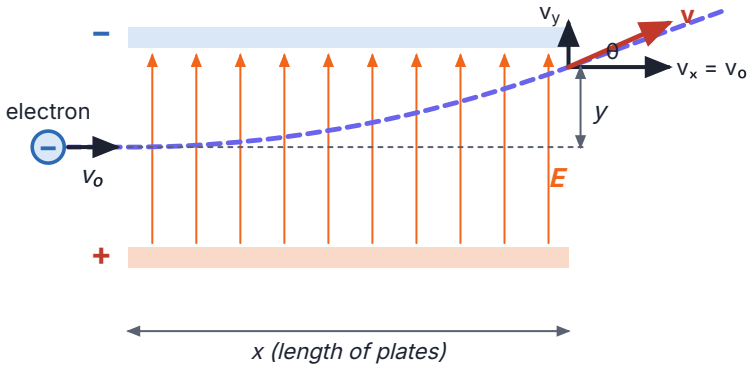


Figure 13.9: an electron enters the field between parallel plates at right angles to the field.

time between the plates: $t = \frac{x}{v_0}$

vertical velocity on leaving: $v_y = at = \frac{qEx}{mv_0}$

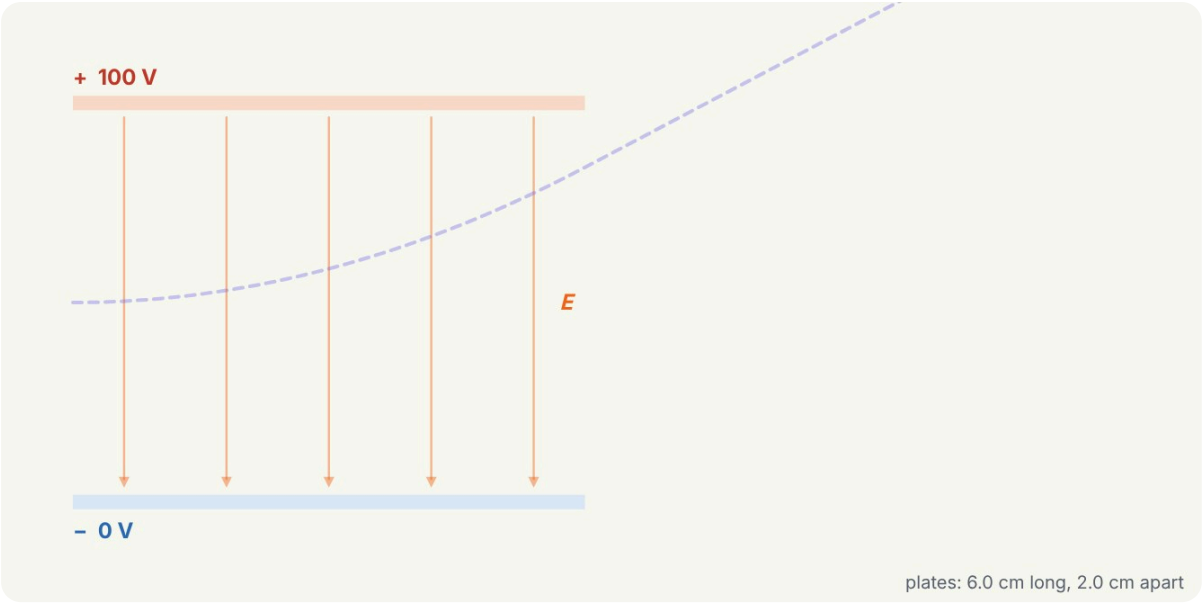
vertical displacement: $y = \frac{1}{2}at^2 = \frac{qEx^2}{2mv_0^2}$ ($y \propto x^2$: a parabola)

speed and direction on leaving: $v = \sqrt{v_x^2 + v_y^2}$, $\theta = \tan^{-1}\frac{v_y}{v_x}$

Simulation

A charge between parallel plates

Choose the particle, then change the voltage across the plates and the entry speed. See when the particle leaves the plates and when it hits one of them.



plates: 6.0 cm long, 2.0 cm apart

field E = V/d	acceleration qE/m	time between plates	vertical deflection y
5.00×10^{3} V m^{-1}	8.78×10^{14} m s^{-2}	4.00×10^{-9} s	7.03 mm

v_y on leaving	angle of deflection θ
3.51×10^{6} m s^{-1}	13.2°

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 13 to use the simulation.

Example 13.2.6

An electron travelling at 1.5×10^7 m s^{-1} enters midway between two parallel plates 60 mm long and 20 mm apart. The field between the plates is 4.0×10^3 V m^{-1}, with the upper plate positive. (a) Sketch the path of the electron between and after the plates. (b) Find the acceleration of the electron. (c) Find the components of its velocity as it leaves the plates. (d) Find the angle of deflection. ($e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg)

Solution

(a) a parabola curving towards the positive plate between the plates, then a straight line after it leaves

(b) $a = \dfrac{eE}{m} = \dfrac{(1.60 \times 10^{-19})(4.0 \times 10^3)}{9.11 \times 10^{-31}} = \mathbf{7.03 \times 10^{14}\ m\ s^{-2}}$, towards the positive plate (opposite to E)

(c) $t = 0.060/1.5 \times 10^7 = 4.0 \times 10^{-9}$ s; $v_x = \mathbf{1.5 \times 10^7\ m\ s^{-1}}$; $v_y = at = (7.03 \times 10^{14})(4.0 \times 10^{-9}) = \mathbf{2.8 \times 10^6\ m\ s^{-1}}$

Check: $y = \tfrac{1}{2}at^2 = 5.6$ mm, less than 10 mm, so it does leave the plates.

(d) $\theta = \tan^{-1}\dfrac{2.8 \times 10^6}{1.5 \times 10^7} = \mathbf{10.6°}$

Example 13.2.7

An electron starts from rest at the negative plate and is accelerated across a potential difference of 500 V. Find its speed when it reaches the positive plate.

Solution

$\tfrac{1}{2}mv^2 = eV \Rightarrow v = \sqrt{\dfrac{2eV}{m}} = \sqrt{\dfrac{2(1.60 \times 10^{-19})(500)}{9.11 \times 10^{-31}}} = \mathbf{1.33 \times 10^7\ m\ s^{-1}}$

13.3 Gauss's law

Electric flux, Φ

through a surface of area A in a uniform field E: **the scalar product of E and the area vector A**. It measures how many field lines pass through the surface.

$$\Phi = \boldsymbol{E} \cdot \boldsymbol{A} = EA\cos\theta$$

θ = angle between E and the **normal** to the surface; unit N m^2 C^{-1} (= V m); scalar

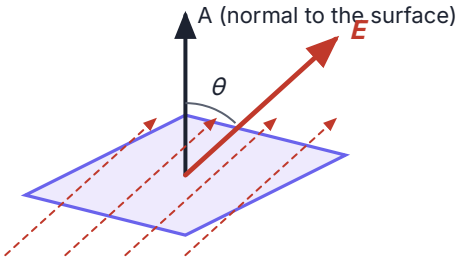


Figure 13.10: the area vector A is normal to the surface.

- θ = 0 (field perpendicular to the surface): Φ = EA, the maximum.
- θ = 90° (field parallel to the surface): Φ = 0, no lines pass through.
- For a **closed** surface the normal points outwards: flux leaving is positive, flux entering is negative.

Gauss's law

The **total electric flux through a closed surface is equal to the net charge enclosed by the surface divided by ε_0**.

$$\Phi = \oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{A} = \frac{Q_{\text{enclosed}}}{\varepsilon_0}$$

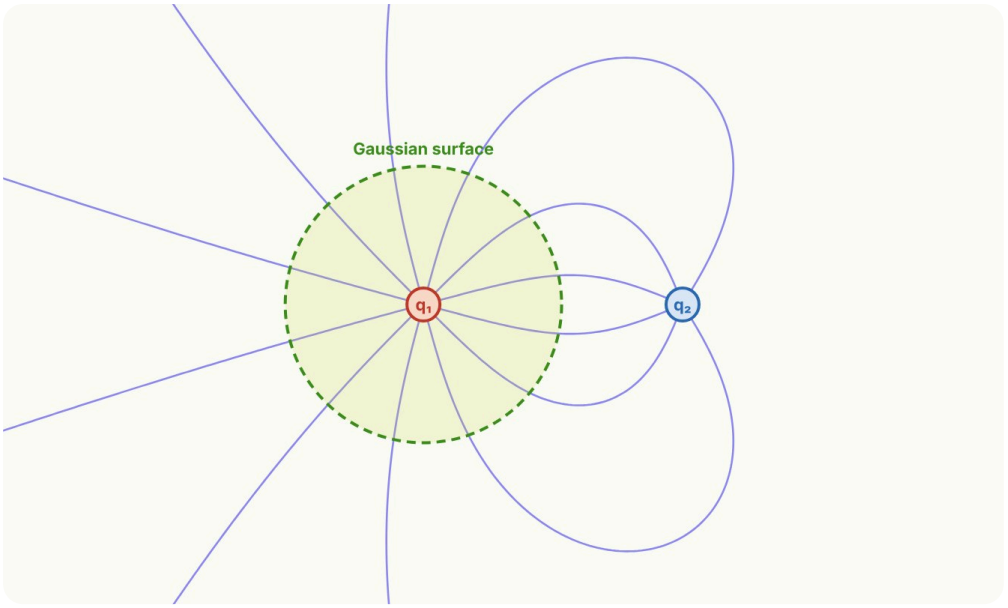
the closed surface is called a Gaussian surface; its shape and size do not matter, only the charge inside

- A charge **outside** the closed surface gives zero net flux: every line that enters also leaves.
- If the net charge inside is zero, the net flux is zero (but the field on the surface need not be zero).

Simulation

Gauss's law

Change the size of the Gaussian sphere and move the second charge in and out. The flux depends only on the charge inside. Then look at a charged conducting sphere: E and V inside and outside.



charge enclosed
+2 nC

flux $\Phi = Q/\varepsilon_0$
226 N m² C⁻¹

q_2 is
outside: its lines go in and out, net flux 0

size of the surface
does not change Φ

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 13 to use the simulation.

Example 13.3.1

A uniform field of 2.0×10^3 N C^{-1} passes through a flat square of side 10 cm. Find the flux through the square when its normal makes (a) 0°, (b) 60°, (c) 90° with the field.

Solution

$A = (0.10)^2 = 0.010$ m^2

(a) $\Phi = (2.0 \times 10^3)(0.010) \cos 0° =$ **20 N m² C⁻¹**

(b) $\Phi = 20 \cos 60° =$ **10 N m² C⁻¹**

(c) $\Phi = 20 \cos 90° =$ **0**

Example 13.3.2

A point charge of +5.0 nC is at the centre of a cube. Find (a) the total flux through the cube, (b) the flux through one face. (c) What is the flux through the cube if the charge is moved outside it?

Solution

(a) $\Phi = \dfrac{Q}{\varepsilon_0} = \dfrac{5.0 \times 10^{-9}}{8.85 \times 10^{-12}} = \mathbf{565\ N\ m^2\ C^{-1}}$

(b) by symmetry the six faces share it equally: 565/6 = **94.2 N m² C⁻¹**

(c) **zero**: no charge is enclosed

Using Gauss's law to find E

Choose a Gaussian surface on which E has the same size everywhere and is perpendicular to the surface. Then $\oint \boldsymbol{E}\cdot d\boldsymbol{A} = E \times$ (area).



Figure 13.11: Gaussian spheres around a point charge and around a charged conducting sphere.

(a) Isolated point charge Q. Take a sphere of radius r centred on the charge. E is radial and the same size at every point on it:

$$E(4\pi r^2) = \frac{Q}{\varepsilon_0} \quad \Rightarrow \quad E = \frac{Q}{4\pi\varepsilon_0 r^2}$$

This is Coulomb's law again.

(b) Isolated charged conducting sphere (radius R, charge Q). The charge spreads over the **outer surface**.

- Outside ($r \geq R$): a Gaussian sphere of radius r encloses Q, so $E = \dfrac{Q}{4\pi\varepsilon_0 r^2}$: the sphere acts like a point charge at its centre.
- Inside ($r < R$): a Gaussian sphere inside encloses no charge, so $E = 0$.
- The potential is the same everywhere inside, equal to the surface value $Q/4\pi\varepsilon_0 R$.



Figure 13.12: E and V against r for a charged conducting sphere.

(c) Uniformly charged conducting plate. The surface charge density is σ = Q/A. Take a small cylinder (pillbox) with one end inside the conductor and the other end, of area A, just outside. Inside the conductor E = 0, and no flux passes through the curved side (E is perpendicular to the surface), so only the outer end counts:

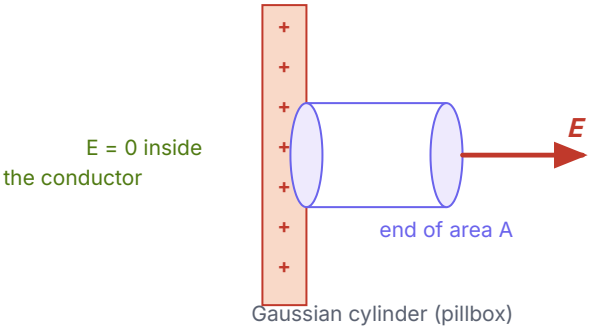


Figure 13.13: a Gaussian pillbox at the surface of a charged conducting plate.

$$EA = \frac{\sigma A}{\varepsilon_0} \quad \Rightarrow \quad E = \frac{\sigma}{\varepsilon_0}$$

The same result, $E = \sigma/\varepsilon_0$, gives the uniform field between two parallel plates with charges $+Q$ and $-Q$. It does not depend on the distance from the plate.

Teacher's note

Why is a car a safe place in a thunderstorm? The metal body is a conductor: the charge stays on its outer surface, and Gauss's law gives E = 0 inside.

Example 13.3.3

A conducting sphere of radius 10 cm carries a charge of +2.0 μC. Find the electric field and the potential at (a) 5.0 cm, (b) 10 cm, (c) 20 cm from its centre.

Solution

(a) inside: $E = 0$; $V = V_{surface} = \dfrac{(9.0 \times 10^9)(2.0 \times 10^{-6})}{0.10} = \mathbf{1.8 \times 10^5\ V}$

(b) $E = \dfrac{(9.0 \times 10^9)(2.0 \times 10^{-6})}{(0.10)^2} = \mathbf{1.8 \times 10^6\ N\ C^{-1}}$; $V = \mathbf{1.8 \times 10^5\ V}$

(c) $E = \dfrac{(9.0 \times 10^9)(2.0 \times 10^{-6})}{(0.20)^2} = \mathbf{4.5 \times 10^5\ N\ C^{-1}}$; $V = \dfrac{(9.0 \times 10^9)(2.0 \times 10^{-6})}{0.20} = \mathbf{9.0 \times 10^4\ V}$

Example 13.3.4

A charge of 4.0 nC is spread evenly over one face of a conducting plate of area 0.020 m^2. Find the electric field just outside the plate.

Solution

$\sigma = Q/A = 4.0 \times 10^{-9}/0.020 = 2.0 \times 10^{-7}\ C\ m^{-2}$

$E = \sigma/\varepsilon_0 = 2.0 \times 10^{-7}/8.85 \times 10^{-12} = \mathbf{2.3 \times 10^4\ N\ C^{-1}}$, perpendicular to the plate

13.4 Electric potential

Electric potential, V

at a point: the **work done to bring a unit positive charge from infinity to that point**. The potential at infinity is zero.

$$V = \frac{Q}{4\pi\varepsilon_0 r} = \frac{kQ}{r}$$

scalar; unit volt, V (= J C^{-1}). Put in the sign of Q: positive near a positive charge, negative near a negative charge.

Derivation

Bring a test charge q_0 from infinity to a point at distance r from a positive charge Q, slowly (no acceleration). The external force must balance the electric force, $F_{ext} = kQq_0/x^2$, pointing towards Q:

$$W = -\int_{\infty}^{r} \frac{kQq_0}{x^2}\,dx = kQq_0\left[\frac{1}{x}\right]_{\infty}^{r} = \frac{kQq_0}{r} \quad \Rightarrow \quad V = \frac{W}{q_0} = \frac{kQ}{r}$$

- For a positive Q, W is positive: work is done **by the external force** against the repulsion.
- For a negative Q, $W = -kQq_0/r$ is negative: the work is done **by the electric force** (the test charge is attracted).



Figure 13.14: the potential against distance for a positive and a negative point charge.

Potential due to several charges

Potential is a scalar: add the potentials from each charge **with their signs**. $V = k(Q_1/r_1 + Q_2/r_2 + \ldots)$

Example 13.4.1

Find the electric potential at a point A, 10 m from a point charge of +5.0 μC, and explain what the answer means.

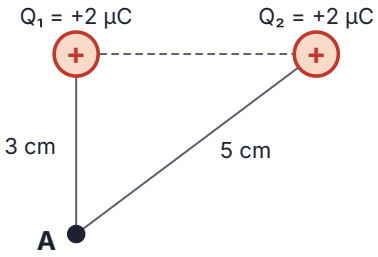
Solution

$$V = \frac{kq}{r} = \frac{(9.0 \times 10^9)(5.0 \times 10^{-6})}{10} = \mathbf{4.5 \times 10^3\ V}$$

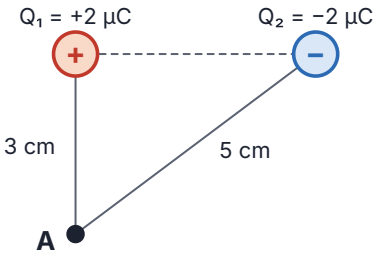
Meaning: 4.5×10^3 J of work is done to bring each coulomb of positive charge from infinity to A.

Example 13.4.2

Find the potential at A in each arrangement.



(a)



(b)

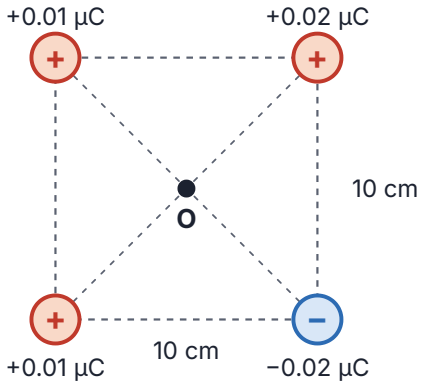
Solution

(a) $V_A = k\left(\frac{Q_1}{r_1} + \frac{Q_2}{r_2}\right) = (9.0 \times 10^9)\left(\frac{2 \times 10^{-6}}{0.03} + \frac{2 \times 10^{-6}}{0.05}\right) = \mathbf{9.6 \times 10^5\ V}$

(b) $V_A = (9.0 \times 10^9)\left(\frac{2 \times 10^{-6}}{0.03} - \frac{2 \times 10^{-6}}{0.05}\right) = \mathbf{2.4 \times 10^5\ V}$ (put in the minus sign of Q_2)

Example 13.4.3

Four charges are at the corners of a square of side 10 cm. Find the potential at the centre O.



Solution

Each charge is $\frac{10}{\sqrt{2}} = 7.07$ cm from O.

$V_O = \frac{k}{r}(\Sigma q) = \frac{9.0 \times 10^9}{0.0707}(0.01 + 0.02 + 0.01 - 0.02) \times 10^{-6} = \mathbf{2.55 \times 10^3\ V}$

Potential difference and work done

Potential difference between A and B

the work done per unit positive charge in moving a charge from B to A (against the field, with no acceleration).

$$V_{AB} = V_A - V_B = \frac{W_{B\to A}}{q} \qquad W_{A\to B} = q(V_B - V_A)$$

W is the work done by an **external force**. The work done by the **electric field** is the negative of this.

- If q and $(V_B - V_A)$ have the same sign, the external force does positive work (the charge is pushed against the field).
- If they have opposite signs, the work done by the external force is negative: the field does the work.

Example 13.4.4

(a) Find the potential at (i) 10 cm, (ii) 50 cm from a point charge of +2 μC. (b) Find the work done to move a charge of +0.05 μC from 50 cm to 10 cm from the point charge. Is the work done by an external force or by the electric field?

Solution

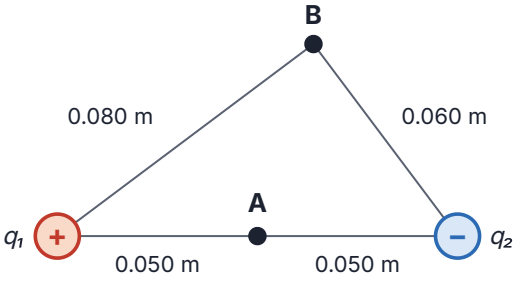
(a)(i) $V = (9.0 \times 10^9)(2 \times 10^{-6})/0.10 =$ **1.8×10^5 V** (ii) $V = (9.0 \times 10^9)(2 \times 10^{-6})/0.50 =$ **3.6×10^4 V**

(b) $W = q(V_{10} - V_{50}) = (0.05 \times 10^{-6})(1.8 \times 10^5 - 3.6 \times 10^4) =$ **7.2×10^{-3} J**

W is positive: the work is done **by an external force against the electric field** (a positive charge is pushed towards a positive charge). The field itself does -7.2×10^{-3} J.

Example 13.4.5

q_1 = +2.40 nC and q_2 = −6.50 nC are 0.100 m apart. A is midway between them; B is 0.080 m from q_1 and 0.060 m from q_2. Find (a) V_A, (b) V_B, (c) the work done by the electric field on a charge of +2.5 nC that moves from B to A.



Solution

(a) $V_A = (9.0 \times 10^9)\left(\frac{2.40 \times 10^{-9}}{0.050} - \frac{6.50 \times 10^{-9}}{0.050}\right) = 432 - 1170 =$ **−738 V**

(b) $V_B = (9.0 \times 10^9)\left(\frac{2.40 \times 10^{-9}}{0.080} - \frac{6.50 \times 10^{-9}}{0.060}\right) = 270 - 975 =$ **−705 V**

(c) work done by the field = $q(V_B - V_A) = (2.5 \times 10^{-9})(-705 + 738) =$ **+8.3 × 10⁻⁸ J**

(The positive charge moves to a lower potential, so the field does positive work. An external force would do -8.3×10^{-8} J.)

Equipotential surfaces

Equipotential surface

a surface on which every point has the same potential.

- **No work** is done in moving a charge along an equipotential surface ($\Delta V = 0$).
- Equipotential surfaces are always **perpendicular to the field lines**.
- Where the equipotentials are closer together, the field is stronger.

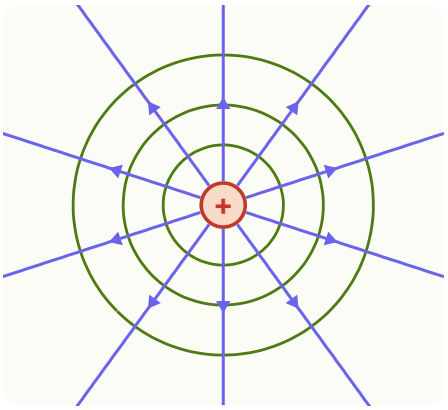


Figure 13.15: point charge: concentric spheres (circles in 2D).

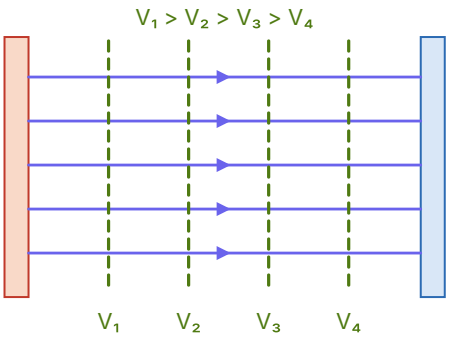


Figure 13.16: uniform field: equally spaced parallel planes.

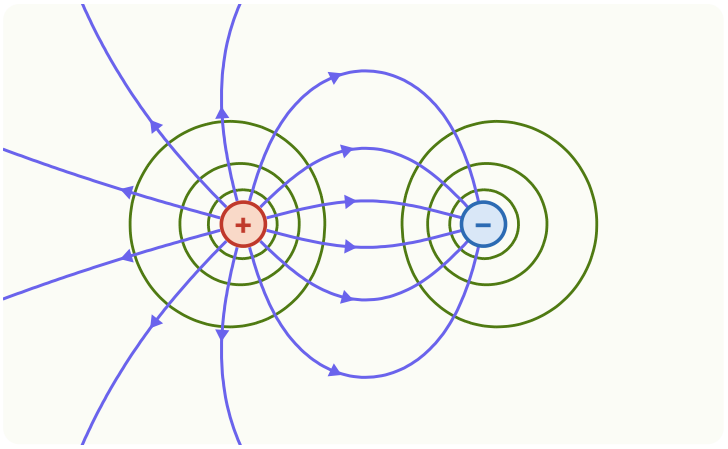


Figure 13.17: electric dipole. Dashed green lines are equipotentials; the middle one (V = 0) is a straight line.

Relation between E and V

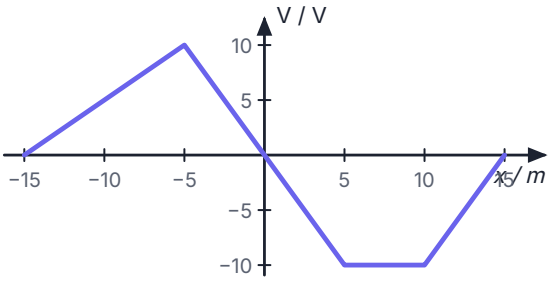
$$E = -\frac{\mathrm{d}V}{\mathrm{d}r} \qquad \text{uniform field: } E = \frac{V}{d}$$

the field strength is the negative of the potential gradient; E points from high potential to low potential

- The steeper the V–r graph, the stronger the field.
- Between parallel plates with p.d. V and separation d, V falls steadily from one plate to the other, so $E = V/d$ is uniform.

Example 13.4.6

The graph shows how the potential V varies with distance x along the x-axis. Sketch the graph of the electric field E against x.



Solution

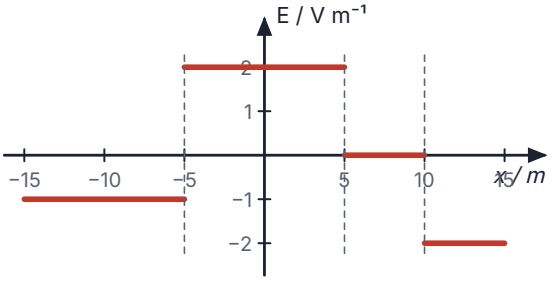
Use $E = -\mathrm{d}V/\mathrm{d}x$ for each straight section:

−15 m to −5 m: gradient = 10/10 = +1 ⇒ E = **−1 V m^{-1}**

−5 m to 5 m: gradient = −20/10 = −2 ⇒ E = **+2 V m^{-1}**

5 m to 10 m: gradient = 0 ⇒ E = **0**

10 m to 15 m: gradient = 10/5 = +2 ⇒ E = **−2 V m^{-1}**



Electric potential energy

Electric potential energy, U

of a charge at a point: the **work done to bring the charge from infinity to that point**.

$$U = qV = \frac{Qq}{4\pi\varepsilon_0 r} \qquad \Delta U = q\Delta V$$

scalar; unit joule. Put in the signs of both charges: $U > 0$ for like charges (repel), $U < 0$ for unlike charges (attract).

For a system of charges, add the potential energy of every pair:

$$U = k\left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}}\right)$$

Example 13.4.7

(a) Find the electric potential energy of a +0.05 μC charge at (i) 10 cm, (ii) 50 cm from a +2 μC point charge. (b) Find the change in potential energy between (i) and (ii).

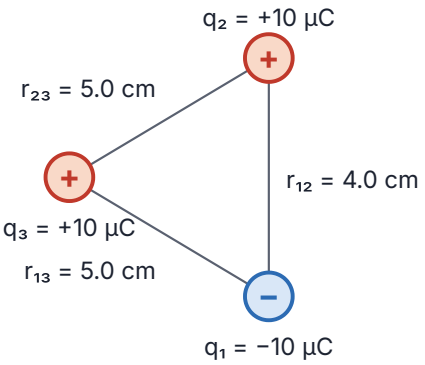
Solution

(a)(i) $U = qV = (0.05 \times 10^{-6})(1.8 \times 10^{5}) =$ **9.0×10^{-3} J** (ii) $U = (0.05 \times 10^{-6})(3.6 \times 10^{4}) =$ **1.8×10^{-3} J**

(b) $\Delta U = 9.0 \times 10^{-3} - 1.8 \times 10^{-3} =$ **7.2×10^{-3} J**: the same as the work done in Example 13.4.4

Example 13.4.8

Find the potential energy of the system of charges shown.



Solution

$$U = (9.0 \times 10^9)\left[\frac{(-10)(10)}{0.040} + \frac{(-10)(10)}{0.050} + \frac{(10)(10)}{0.050}\right] \times 10^{-12}$$

$$= (9.0 \times 10^{-3})(-2500 - 2000 + 2000) = \mathbf{-22.5\ J}$$

Negative: work would have to be done to pull the charges apart to infinity.

Exercise: electric potential

1. Two parallel plates 5.0 mm apart have a potential difference of 100 V. Find the electric field between them and the force on an electron there.

   Answer

   $E = 100/5.0 \times 10^{-3} = 2.0 \times 10^4$ V m^{-1}; $F = eE = 3.2 \times 10^{-15}$ N, towards the positive plate

   FULL SOLUTION

   E = V/d = 100 / (5.0×10^{-3})

   = **2.0×10^4 V m^{-1}**

   F = eE = $(1.60 \times 10^{-19})(2.0 \times 10^4)$

   = **3.2×10^{-15} N, towards the positive plate** (the electron is negative)

2. How much work must an external force do to bring two protons from infinity to 1.0×10^{-14} m apart?

   Answer

   $W = ke^2/r = (9.0 \times 10^9)(1.60 \times 10^{-19})^2/1.0 \times 10^{-14} = 2.3 \times 10^{-14}$ J

   FULL SOLUTION

   The work done equals the electric potential energy of the pair: W = U = ke^2/r

$W = (9.0 \times 10^{9})(1.60 \times 10^{-19})^{2} / (1.0 \times 10^{-14})$

$= \mathbf{2.3 \times 10^{-14}\ J}$

3. Explain why no work is done when a charge moves along an equipotential surface, and why the field lines must cross the equipotentials at right angles.

Answer

$W = q\Delta V$ and $\Delta V = 0$ along the surface. If E had a component along the surface, a force would act along it and work would be done, so E must be perpendicular to the surface.

FULL SOLUTION

Work done moving charge q from A to B: $W = q\Delta V = q(V_B - V_A)$.

All points on an equipotential surface have the same potential, so $\Delta V = 0$ and **$W = 0$**.

If E had a component along the surface, a force qE would act along the surface and work would be done when the charge moves along it, which contradicts W = 0.

So **E has no component along the surface: field lines cross equipotentials at right angles**.

Summary

Quantity	Point charge Q	Scalar or vector	Unit
Force on q	$F = kQq/r^2$	vector	N
Field strength	$E = kQ/r^2 = F/q$	vector	N C^{-1} or V m^{-1}
Potential	$V = kQ/r$	scalar	V (J C^{-1})
Potential energy of q	$U = qV = kQq/r$	scalar	J
Flux through a closed surface	$\Phi = Q_{enclosed}/\varepsilon_0$	scalar	N m^2 C^{-1}
Link	$E = -\mathrm{d}V/\mathrm{d}r$; uniform field $E = V/d$		

Real-World Physics: the Burj Khalifa lightning conductor

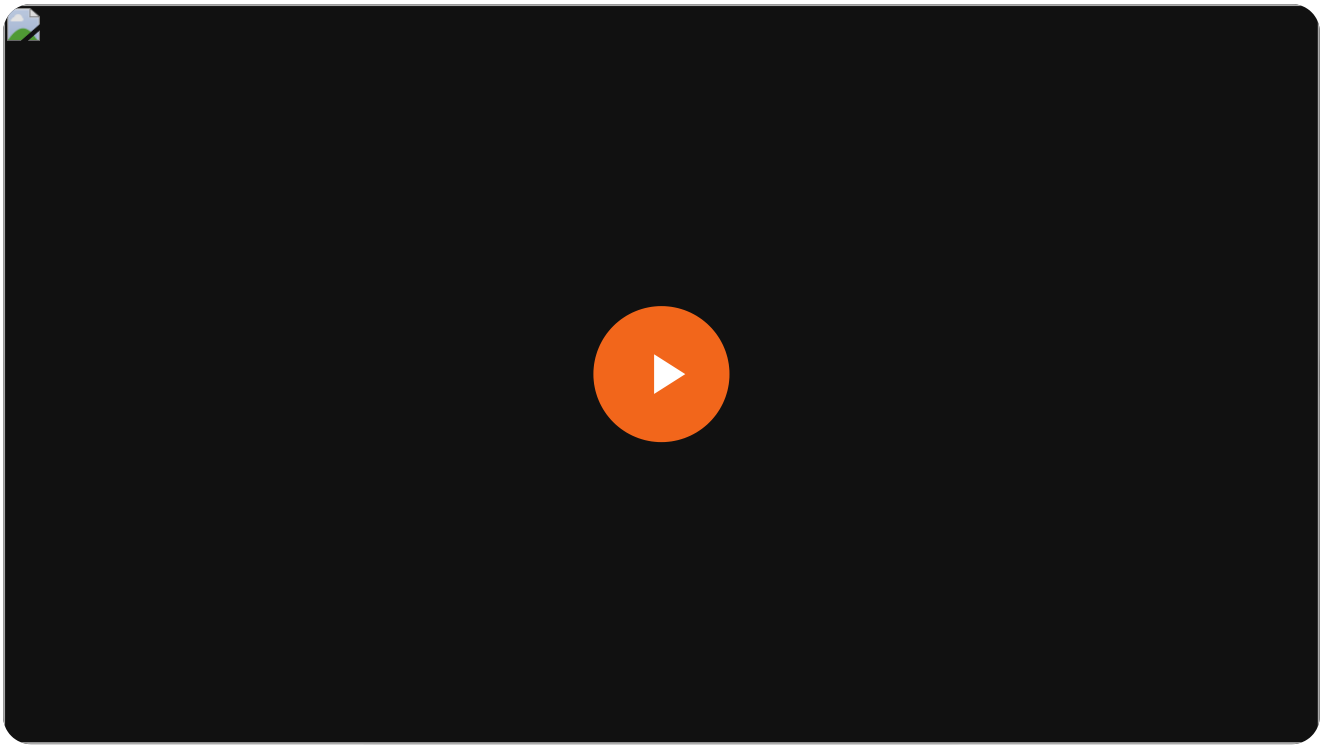
REAL-WORLD PHYSICS

Dubai · Richard Hammond's BIG

Does the Burj Khalifa collect lightning for electricity?

Being the tallest structure for miles, the Burj Khalifa is struck by lightning during thunderstorms. In *Richard Hammond's BIG*, Richard climbs to its lightning conductor at the very top.

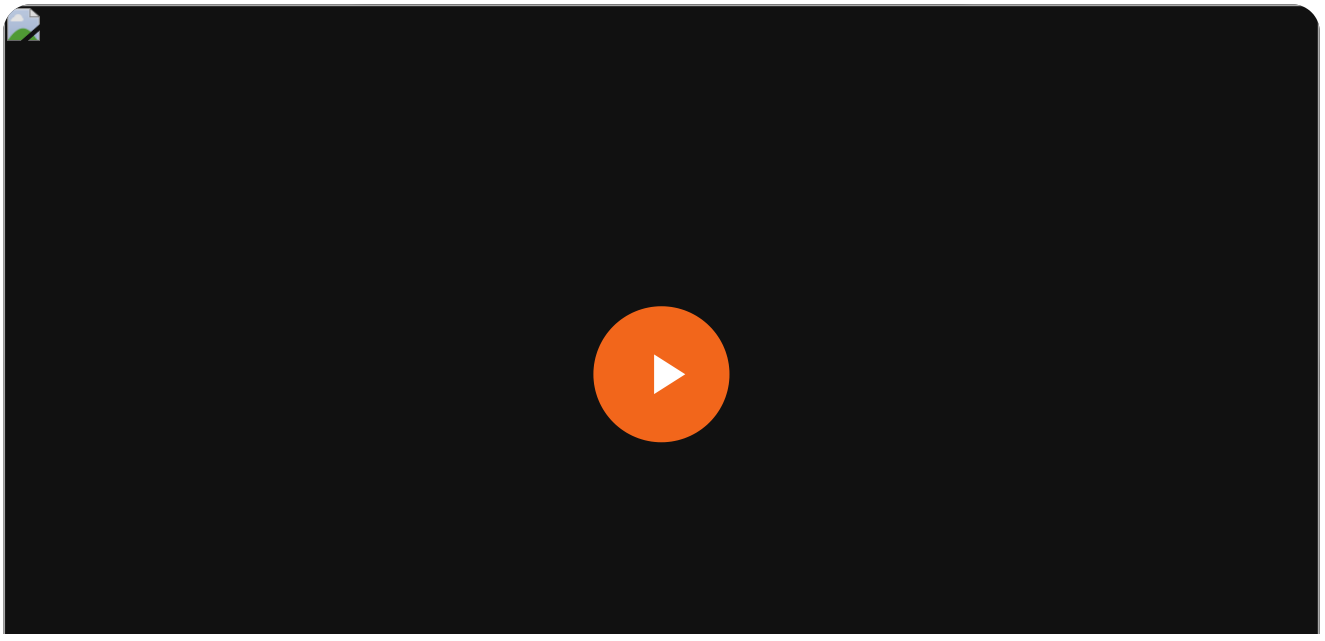
A common misconception: the tower does **not** collect lightning to use as electricity. The lightning protection system safely carries the charge to earth. The tower's steel frame also acts as a **Faraday cage**, so the current flows around the outside and the people inside are safe. No one has yet found a practical way to store the energy of lightning.



How The Burj Khalifa Is Dubai's Lightning Rod · Richard Hammond's BIG Open in YouTube ↗

While you watch: notice the shape of the conductor at the top, and where the charge goes after a strike.

ALSO WATCH





Richard Hammond inside a car struck by lightning (Top Gear) · BBC Studios

Open in YouTube ↗

While you watch: Richard sits inside a car while it is hit by an artificial lightning strike. Watch where the discharge travels, and think about which part of the car protects him.

The physics

- On a charged conductor, charge collects most densely at **sharp points**. The electric field there is very strong and can ionise the air, giving lightning an easy path to the conductor.
- The conductor gives a **low-resistance path** to earth, so charge flows safely into the ground instead of through the building.
- Inside a closed conductor the electric field is zero, and charge stays on the outer surface (Gauss's law). This is why a steel-framed building protects the people inside.
- A strike carries about 1×10^9 J, but it lasts only tens of microseconds and its time and place cannot be predicted, so it cannot practically be stored.

Quick check 1

Why does a lightning conductor have a sharp, pointed tip?

A Sharp points have a lower electric potential than flat surfaces.

B Charge density is greatest at sharp points, so the electric field is strongest there and the air is easily ionised.

C Sharp points have a smaller resistance than the rest of the conductor.

D Sharp points repel lightning away from the building.

Quick check 2

Why are people inside a steel-framed building safe when lightning strikes it?

A The electric field inside a closed conductor is zero, and the charge stays on its outer surface.

B Steel is an electrical insulator.

C The lightning energy is stored in the steel frame.

D The electric potential inside the building is always zero.

Quick check 3

In the Top Gear clip, Richard Hammond is unharmed when lightning strikes the car he is sitting in. What is the main reason?

A The rubber tyres insulate the car from the ground.

B The metal body of the car conducts the charge around the outside to earth, so there is no electric field inside.

C The glass windows absorb the energy of the lightning.

D The car is moving, so the lightning cannot reach the inside.

Quick check 4

A lightning strike carries 1.0×10^9 J. A home uses 30 kWh of electrical energy per day. For how long could one strike supply the home, if all the energy could be stored?

A 0.3 days

B 9.3 days

C 33 days

D 9.3 years

Quick check 5

A lightning strike has an average current of 30 kA for 50 µs. How much charge flows to earth?

A 1.5 C

B 600 C

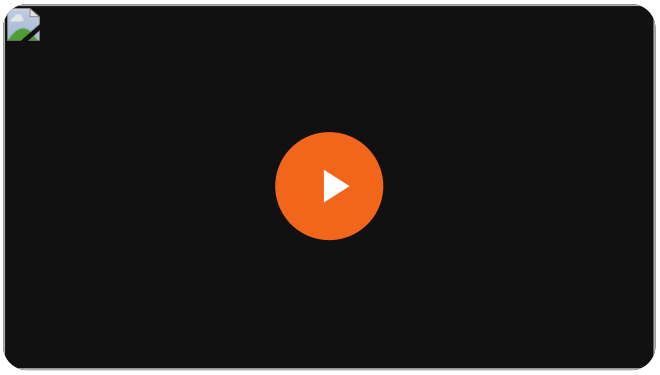
C 1.5×10^{-3} C

D 6.0×10^8 C

Sources: *Richard Hammond's BIG*, Episode 7 "Tallest Building On Earth" (Discovery, 2020); *Top Gear* (BBC Studios); The National (UAE); UNSW Sydney. Preview the clip before class. youtube.com/watch?v=E0RdTXJaqJs

Physics in Real Life

Real-world videos for this chapter. Tap a picture to play it here.



LIVE DEMONSTRATION

Demonstration of How Lightning Rods Work

Open in YouTube ↗

How a lightning rod works

A Van de Graaff generator shows charge gathering on a pointed conductor and the strong electric field that lets charge leak away safely.